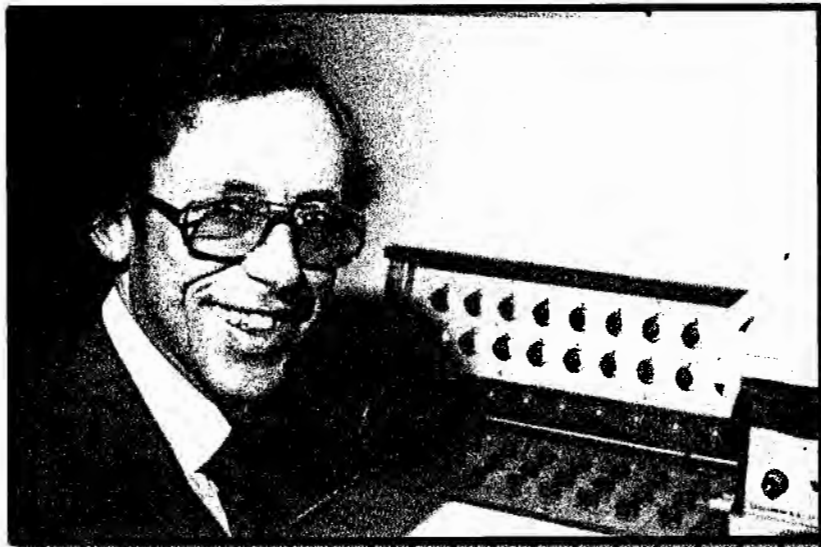


R Brown
15 June 2023

Psychology and
Neuroscience 47th Annual
Graham Goddard In-House
Conference



June 16, 2023

Life Sciences Centre Room 4260-5260

TALK ABSTRACTS

(In Alphabetical Order of Presenter)

Badcock-Parks, Broderick

TALK TITLE: Growth of vocal control nucleus Area X precedes gonadal recrudescence in wild male black-capped chickadees (*Poecile atricapillus*)

AUTHORS: Broderick M. B. Parks^{1*}, Kyle McVea¹, Leslie S. Phillmore¹

¹Department of Psychology and Neuroscience, Dalhousie University, Halifax NS

ABSTRACT: In temperate songbirds, spring breeding season brings about changes in behaviour, physiology, and neuroanatomy; these changes are initiated by a longer photoperiod (day length). It is believed that breeding-associated neuroanatomical changes, including growth of nuclei of the vocal control system, occur downstream of photoperiod-induced gonadal recrudescence, however, evidence from wild birds is mixed. Here, we captured wild black-capped chickadees ($N=18$ adult males) in spring (March-April), summer (August-September), and winter (December-January). We quantified the volume of two vocal control nuclei: Area X (critical for vocal learning) and HVC (critical for production). Most adult males (6/8) captured in spring had small testes and low levels of circulating androgens, indicating these birds were not yet in full breeding condition. However, spring males still had a significantly larger Area X (i.e., sizes expected in breeding season) than winter males, while HVC did not differ across groups. These data suggest that the growth of Area X, but not HVC, occurs upstream (or independently) of gonadal recrudescence, and may also reflect differential temporal and behavioural demands on the songbird vocal control system in advance of breeding.

Barresi, John

TALK TITLE: Kafka's *Metamorphosis*

AUTHORS: John Barresi

ABSTRACT: The hero of Franz Kafka's story, *The Metamorphosis*, wakes up one morning as a huge beetle-like bug. Though he retains memories from his past life, his consciousness adapts to this radically different body until he dies from an injury caused by an apple his father threw at him. What are we to make of this transformation in self-identity? I will describe the distinct roles that first- and second/third-person perspectives have in one's experience of self and Kafka's failure at integrating these two perspectives in early development. I base my account on collaborative work with Chris Moore and works by other authors. It concludes with a discussion of Heinz Kohut's theory of how the 'modern' self relies on a vertical splintering of embodied being-in-the-world rather than a horizontal split between a conscious mind and unconscious embodiment found in theories from Descartes to Freud.

Blais, Julie

TALK TITLE: Assessing the Relationship between Antisocial Personality Traits and Cyberstalking in a large sample of Canadian Adults

AUTHORS: Julie Blais, Charlotte A. Aelick, J. Michelle Scully, & Scott Pruyssers

ABSTRACT: Cyberstalking is defined as the repeated pursuit of an individual using the Internet. Several studies have

identified the Dark Triad (psychopathy, Machiavellianism, and narcissism) as related to cyberstalking behaviours; however, this research has tended to use less-than-ideal measures of the traits. The purpose of the current study was to assess the associations between each Dark Triad construct among a large sample of people currently living in Canada ($N = 1,725$) while addressing several earlier limitations. After controlling for age and sex, only the antisocial facet of psychopathy (early and repeated antisocial behaviour; $OR = 1.04$, 95% CI [1.02, 1.06]) and grandiose narcissism (inflated self-esteem and antagonism; $OR = 1.01$, 95% CI [1.00, 1.02]) were significantly related to cyberstalking behaviour within the past 12 months. We will discuss how to improve Dark Triad research moving forward and what the results mean for identifying dynamic risk factors that can be targeted in evidence-based interventions.

Dockrill, Mya

TALK TITLE: Barriers and Facilitators to the *Better Nights, Better Days* during COVID-19 Program for Parents and Healthcare Providers: A Qualitative Perspective

AUTHORS: Mya Dockrill, BSc (Presenting Author), Elizabeth Keys, PhD, Amber Myatt, Penny Corkum, PhD

ABSTRACT: Due to concerns surrounding children's mental health and sleep during the COVID-19 pandemic, *Better Nights, Better Days* (BNBD), was modified to support parents of children with sleep difficulties during the pandemic. Using virtual, semi-structured interviews, this study sought to understand the barriers and facilitators that parents ($n = 34$) and healthcare providers (HCPs) ($n = 15$) faced when completing and referring to the BNBD-COVID-19 program. Most parents and HCPs believed the program was effective in improving children's sleep, sleep knowledge, and/or daytime behaviour. Half of the parents reported that the pandemic made the program harder to implement, and most parents mentioned that time commitment and/or other external circumstances impeded their ability to complete the program. Interview feedback will be used to improve the program to better support children with sleep difficulties, as well as to improve HCP experiences when referring to the program in the context of a crisis.

Elliot, Laura

TALK TITLE: Lexical Decisions in Ortho-semantic Learning: An ERP Study of Grade 3 Children in English and French Immersion

AUTHORS: Laura Elliott (Supervised by: Aaron Newman)

ABSTRACT: Children in French immersion (FI) experience a lag in English literacy skills through the early grades of elementary school, until English instruction is introduced into the curriculum. Here we recruited third grade English and FI students to complete a novel-word learning paradigm, and a lexical decision task during EEG recording, to compare N170 print and lexical tuning effects between the two groups. The N170 print and lexical tuning effects are measured using event-related brain potentials (ERPs) and have been shown to relate to behavioural measures of reading development. We hypothesized FI students would demonstrate bilateral N170 print tuning and no lexical tuning, while English students would demonstrate more left-lateralized print and lexical tuning of the N170, and higher reading proficiency. Findings demonstrated English and FI students performed similarly on behavioural assessments of reading. However, while both groups demonstrated print tuning effects, lexical tuning was only observed for English students. These findings demonstrate measurable differences in neural activity during reading between English and FI students, that do not necessarily indicate a lag in reading skill for FI students.

Foxall, Victoria

TALK TITLE: Sleep for Health in Hospital and at Home

AUTHORS: Victoria Foxall, Dr. Megan Thomas

ABSTRACT: Healthy sleep during hospitalization supports recovery, pain management, and resilience. Many pediatric

patients and their accompanying guardians experience poor sleep in hospital due to family- and environment-related factors. Patients (2 to 16) and their accompanying guardians staying at least 2 nights in the PMU were invited to participate in a mixed-methods exploratory study comparing sleep in hospital to home. This included an optional semi-structured interview exploring sleep experiences, and disruptors and promoters of sleep. Interviews were audio-recorded, transcribed verbatim, subjected to thematic analysis guided by the Appreciative Inquiry framework, and compared to themes from a previous healthcare provider (HCP) survey. Findings showed a shared understanding between HCP and parents regarding many of the sleep disruptors, possible solutions and examples of supportive practices. COVID-19 was noted to be an additional challenge. Parents provided novel suggestions to encourage sleep and wellbeing and reported positive changes to their sleep at home following participation.

Garden, Jessica

TALK TITLE: Pavlovian conditioned odour memory in the *Neurexin1* +/- mouse model of autism spectrum disorder

AUTHORS: Jessica F. Garden, Ian C. G. Weaver, Richard. E. Brown. Department of Psychology and Neuroscience, Dalhousie University, Halifax, N.S. Canada B3H 4R2

ABSTRACT: Neurexins (NRXNs) are presynaptic cell-adhesion molecules that bind with postsynaptic ligands to regulate synaptic development and function. Altered *Nrxn* gene expression is linked to neurodevelopmental disorders, such as autism spectrum disorder (ASD), in which learning and memory are impaired. The *Nrxn1* +/- transgenic mouse model, with the loss of one *Nrxn1* allele, is a novel model of ASD. Male and female *Nrxn1* +/- and WT (*Nrxn1* +/+; C57BL/6J) mice at 90-120 days of age were conditioned to dig in an odor pot to receive a sugar reward over 8 trials in a 1-day Pavlovian conditioning protocol. Their short- and long-term olfactory memory was compared with that of naïve mice receiving no odour conditioning. Using the time spent digging in the odour pot as a measure of olfactory memory, we hypothesised that *Nrxn1* +/- mice would have impaired long-term memory, however, they did not differ from the WT mice during the training day nor during 24-hour or 7-day memory tests. The Pavlovian conditioned mice, however, showed significantly more digging in the odour pot than naïve mice during both the learning and memory trials, indicating that digging behaviour accurately reflects olfactory learning and memory in this task. Overall, the results indicate that *Nrxn1* +/- mice do not exhibit impaired olfactory learning or memory based on this 1-day Pavlovian conditioning odour paradigm.

Gaudreau, Sierra

TALK TITLE: The effect of sensory feedback modality on workload and performance in a remote teleoperation task

AUTHORS: Sierra Gaudreau, Chris Holland, Lucas Wan, Dr. Ya-Jun Pan, Dr. Heather Neyedli

ABSTRACT: Combined visual-haptic feedback improves robot teleoperation without an increase in user workload¹. However, these benefits have only been seen when each modality of feedback is either present or absent and have not been investigated in a graded manner that is more reflective of the settings in which this technology is used. Therefore, this study sought to investigate the impact of imperfect visual and/or haptic feedback on teleoperation and perceived workload. 25 participants navigated a virtual robot using a haptic device in four different conditions with either ideal or impeded visual and haptic feedback. Time to complete each trial and perceived workload were recorded. Only impeded visual feedback significantly impacted navigational ability and perceived workload, haptic feedback did not improve performance. These findings prompt further investigation into the user benefits of cross modal designs to examine if this type of system design is only beneficial to users in high workload tasks.

Goruk, KJ

TALK TITLE: Sexual and Gender/Sexuality-Based Harassment in Canada: Prevalence and the Identification of Risk Factors

AUTHORS: KJ Goruk, Julie Blais, Jennifer McArthur

ABSTRACT: Sexual and gender/sexuality-based harassment are common and damaging forms of non-contact violence that can occur in-person or online. The purpose of this study was to determine the prevalence of both types of harassment in Canada and to identify whether established risk factors for offending behaviours (i.e., younger age, general antisociality, sexual deviancy), are predictive of each type of perpetration. Participants ($N = 1,151$) were recruited via Qualtrics' research panels and completed measures assessing various risk factors and were asked whether they had perpetrated both types of harassment in-person and/or online in the preceding twelve months. Overall, 5.2% of the sample reported perpetrating both types of harassment, 4.0% reported perpetrating only sexual harassment, and 4.8% reported perpetrating only gender/sexuality-based harassment within the last year. Salient predictors for both types of harassment were being younger in age, being a man, and scoring higher on telephone scatologia. Hypersexuality and hostile aggression were important for predicting sexual harassment only, and alcohol use and the affective facet of psychopathy were important for predicting gender/sexuality-based harassment only. Results suggest that generalized intervention programs for those who perpetrate both types of harassment may be effective, however, given inconsistencies in predictors for each behaviour, programs should be malleable to each individual's primary risk factors.

Henneberry, Mark

TALK TITLE: Abnormal Visual Experience Attenuates the Capacity for Neural Plasticity

AUTHORS: Mark Henneberry, Joseph Elgallad, Seth Smith, and Kevin Duffy

ABSTRACT: Amblyopia is a debilitating visual impairment caused by an obstruction of normal binocular vision. Conventional treatment for amblyopia involves occluding the fellow eye to allow neural connections serving the amblyopic eye to recover. The effectiveness of this therapy is enabled by a high capacity for plasticity early in postnatal life. It has been suggested that recurrent abnormal visual experience exhausts the potential for neural plasticity in the visual system. In this study, we assessed the impact of an early and transient monocular deprivation (MD) on the effects of a subsequent period of MD to the fellow eye. The atrophy of soma size and loss of neurofilament typical of MD was reduced in magnitude in animals that experienced a prior MD. These findings suggest that cumulative obstruction of normal vision attenuates the capacity of neural, and may impact the successful treatment of amblyopia.

Herbison, Ryan

TALK TITLE: Mechanisms behind parasitoid wasp induced behavioral changes in the tobacco hornworm caterpillar

AUTHORS: Ryan Herbison, Nicholas Raun, Laura McMillan, Dylan Miller, Shelley Adamo

ABSTRACT: Parasites, capable of altering a host's behaviour to facilitate their own life cycle, provide a rare window into the complex physiology of the brain and how to change it. *Cotesia congregata*, a parasitoid wasp, appears to significantly affect the behaviour of the tobacco hornworm caterpillar *Manduca sexta* upon emergence from its body wall, suppressing its development, feeding and motivation, culminating in a body guarding behaviour. How these behaviours are induced by the parasitoid wasp is unknown. Elucidating the mechanisms the parasite uses to change behaviour, in this system and others, could be foundational to new methods for interacting physiology, therapeutically or otherwise. In this study we contrasted the transcriptome of parasitized vs. non-parasitized caterpillar brains at multiple time points throughout infection, from prior emergence to 3 days after. At emergence, we found many parasite transcripts present in the caterpillars brain, with functions around synapses, vesicle transport, neuropeptide processing and gene expression.

Hill, Taylor

TALK TITLE: Crafting happiness from everyday life: Personality, personal projects, basic psychological need satisfaction, and well-being

AUTHORS: Taylor G. Hill, Emma C. Coughlan, & Sean P. Mackinnon

ABSTRACT: Feeling competent, related, and autonomous promotes well-being through satisfying basic psychological needs, according to self-determination theory's basic psychological need satisfaction mini-theory. Personal projects are personally relevant goal-directed activities that take place over an extended period. The quality of life elicited from pursuing personal projects depends on the degree to which projects provide a sense of relatedness, competence, and autonomy. We expected that, when controlling for perfectionistic standards and discrepancies, achievement striving would lead to the pursuit of projects that provide a sense of competence, autonomy, and relatedness, which in turn leads to well-being. The sample (N = 327) was composed of students and the general adult population who provided information on positive mental health, passion, zest for life, life purpose, personality, basic psychological need satisfaction, and personal projects. We used a cross-sectional survey design and tested hypotheses with twelve serial mediation models. Achievement striving and personal standards was positively associated with competence, which in turn predicted well-being, though effect sizes tended to be larger for personal standards. Achievement striving and discrepancies generally did not contribute to change in well-being through relatedness or autonomy after controlling for personal standards. Those oriented toward achievement motivation are likely to feel competent in their pursuits (personal projects), which in turn promotes well-being. Pursuing personal projects that suit one's personality (i.e., make it more likely to meet basic psychological needs) may be a tool to boost well-being.

Holland, Chris

TALK TITLE: Trust in Automated Systems is Influenced Differently by Increasing and Decreasing Automation Reliability

AUTHORS: Benjamin S. P. Rittenberg^{1*}, Christopher W. Holland^{2*}, Grace E. Barnhart¹, Sierra M. Gaudreau² Heather F. Neyedli¹

¹School of Health and Human Performance, Dalhousie University, Halifax, Nova Scotia, Canada

²Department of Psychology and Neuroscience, Dalhousie University, Halifax, Nova Scotia, Canada

*These authors contributed equally to this work.

ABSTRACT: The reliability level of automation can affect users trust and performance, however, the temporal dynamics of trust in relation to changing reliability levels remains relatively unexplored. The purpose of the study is to determine how user trust evolves over time when automation reliability increases or decreases. Three groups of participants performed a binary decision task with the help of an automated aid. Group 1 used automation that increased from 50%-100% reliability, group 2 increased from 70%-100% reliability, and group 3 decreased 100%-50% reliability. Self-confidence and trust were measured at regular intervals. Trust in automation only increased in the 70%-100% reliability level and participants' trust was better calibrated to the task when they had higher self-confidence. Human-automation performance tracked with automation reliability level. The results indicate that users need to interact with moderately reliable automation to improve their trust in automation.

Hurst, Austin

TALK TITLE: Understanding The Effects of Motor Imagery on Effector Mapping

AUTHORS: Austin J. Hurst & Shaun G. Boe

ABSTRACT: Motor imagery has been found to improve motor performance in a wide range of contexts, from laboratory

tasks to professional athletics. Despite its effectiveness, however, the cognitive mechanisms behind motor imagery remain poorly understood. We designed a study to test whether motor imagery rehearses effector mapping, which occurs late in the motor process. Participants performed three blocks of a fine-motor pointing task using a joystick: a brief familiarization block, a training block consisting of either physical practice, motor imagery, or a control task, and a final test block where the joystick's y-axis was inverted. The more the original effector mapping was rehearsed in the first two blocks, the larger the performance impairment we would expect in the final block relative to the first. Preliminary analyses indicate that the MI group was less impaired by the inverted axis than the PP group, suggesting a notable neural divergence between motor imagery and physical practice with important practical applications.

Klein, Raymond

TALK TITLE: Mental effort, pupil size, task difficulty and endogenous spatial attention

AUTHORS: Raymond M Klein, Dawson Sutherland, Austin Hurst

ABSTRACT: Does mental effort, as measured by pupil dilation, adjust for task difficulty (Kahneman & Beatty, 1966) and matter for task performance in an endogenous cuing task (Posner, Snyder & Davidson, 1980)? We monitored eye movements and measured pupil size while participants responded to peripherally presented targets preceded by informative central cues. The colour of the central fixation stimulus provided information about whether the task would be easy (simple detection) or harder (simple detection followed by the report of the colour of the target) and none of the events during a trial would involve the kind of change in luminance that might confound our use of pupil size as a measure of mental effort. Extending Kahneman's finding from memory to perception, the hard task elicited a greater investment of mental effort (greater pupil dilation) than the easy task. Cuing effects, which decreased monotonically with practice, were relatively unaffected by mental effort but early in practice larger pupils (greater effort) resulted in reduced costs on invalid trials. Implications of these findings will be discussed.

Lewis, Ryan

TALK TITLE: Exploring the Role of Attentional Withdrawal in Item-Method Directed Forgetting Through Visual Search

AUTHORS: Ryan Lewis, Tracy L. Taylor

ABSTRACT: We embedded a serial and pop-out visual search into the study phase of an item method-directed forgetting paradigm to assess attentional availability following instructions to remember (R) or forget (F). Study words were presented to the left or right of a central fixation (Experiments 1-2) or at center (Experiments 3-5), followed by a visual (Experiments 1-4) or auditory (Experiment 5) memory instruction, and then by the search display. Although the evidence did not support a differential withdrawal of attention following F compared to R instructions, Experiment 5 replicated slower response times to visual targets that followed F compared to R instructions (e.g. Fawcett and Taylor, 2008; 2012) - consistent with the view that intentional forgetting engages an active cognitive process. We speculate that this active process might be a non-attentional inhibitory mechanism that is employed to further reduce the encoding of F items alongside selective rehearsal of R over F item.

Mackinnon, Sean

TALK TITLE: Understanding fear in statistics classrooms: The relationship between perfectionism, negative feedback, and statistics anxiety

AUTHORS: Sean P. Mackinnon, Sean Alexander, Robert A. Cribbie, Gordon L. Flett, Taylor G. Hill & Rebecca Workye

ABSTRACT: Anxiety is a problem for statistics education, with many students experiencing statistics-related fears. In three studies, we examine the relationship between perfectionistic personality, negative feedback (i.e., grades) and statistics anxiety.

Study 1 (N=200) validated stimuli to be used as experimental manipulations for negative feedback. We generated two 5-item quizzes (easy & hard) from 30 multiple choice questions by assessing percentages of students answering questions correctly.

Study 2 (N=329) used a two-group, pre-post, between-subjects experimental design. Students completed measures of self-critical perfectionism and state anxiety and were randomized to the easy or hard statistics test condition. Using moderated multiple regression, we found an interaction effect where perfectionistic students had more negative reactions to failing a statistics quiz.

Study 3 examined correlational data from York and Dalhousie (N=271) students currently taking a statistics class. Results show that self-critical perfectionism is positively correlated with domain-specific statistics anxiety and that GPA is negatively related to computational self-concept.

Findings suggest perfectionistic students find statistics education anxiety-provoking. Educators may want to address perfectionistic attitudes in statistics classrooms to facilitate lowered anxiety and reactivity to mistakes.

MacPhee, Catrina

TALK TITLE: Development of a new treatment protocol for spatial neglect post-stroke: Protocol to test feasibility and effectiveness

AUTHORS: Catrina MacPhee, Noémie Bergeron-Germain, Jasmine R. Aziz, Gina Vaillancourt, Teagan Milligan, Gail Eskes & Anne Sophie Champod

ABSTRACT: Spatial neglect is a common attentional condition post-stroke linked to poor rehabilitation outcomes. We have validated a game-like and computerized prism adaptation procedure (Peg-the-Mole, PTM) for the treatment of spatial neglect. Our goal now is to extend this treatment to iPad administration and to test the feasibility and effectiveness of the protocol in stroke inpatient and home settings. Forty right-hemisphere stroke patients experiencing mild to severe symptoms of neglect will be randomized to two groups: Use of PTM with 15° prism goggles (intervention) or 5° prism goggles (placebo control). Patients will complete ten treatment sessions within two weeks with the PTM prism adaptation procedure for 15 minutes per day. Blinded assessments will take place immediately before and after treatment and at one month follow-up. The PTM procedure has the potential to improve patient recovery and quality of life after stroke due to its gamified and portable features that promote treatment accessibility and adherence. The results will help determine if the treatment can be used successfully in inpatient and outpatient settings and who is likely to benefit.

Maxwell, Amy

TALK TITLE: Can a Tired Brain be Trained?

AUTHORS: Amy E. Maxwell, Richard S. Drake, Gail A. Eskes

ABSTRACT: Working memory, the ability to hold and manipulate information, is a key determinant of many higher-order cognitive functions. Research has demonstrated that adaptive working memory training paradigms, namely the *n*-back task, are more beneficial when paired with mindfulness meditation (MM), then when used alone. The underlying mechanisms of the relationship between MM and working memory are not entirely established. MM is, however, associated with a reduction in mental fatigue. The primary objective of this study was to examine hypothesis that mental fatigue acts as a partial mediator between MM and working memory training. Sixteen healthy adults listened to an audio tape of either MM or a Grimm's fairy-tale (active control), were assessed on their mental fatigue, and then completed *n*-back training. Pre- and post-intervention, participants were tested on measures of working memory, and mindfulness skills. Our results revealed that while participants in the MM group had decreased scores of mental fatigue, there was no difference in measures of working memory between groups, suggesting that a reduction in mental fatigue may not be the mechanism by which MM produces gains in working memory capacity.

McArthur, Jennifer

TALK TITLE: Coerced sexting in committed relationships: Insights from a socio-cognitive perspective

AUTHORS: Jennifer McArthur, Julie Blais, and Marguerite Ternes

ABSTRACT: Coerced sexting is recognized as a form of sexual violence; however, little attention has been afforded to sexting between partners in romantic relationships. The current study investigated young adults' attitudes and social perceptions of coerced sexting using the theory of planned behaviour (TPB) as a framework. Young adults ($N=401$) completed a survey assessing the central constructs of the TPB (i.e., attitudes, subjective norms, perceived behavioral control), past behaviour, and intentions to pressure a partner to sext. Attitudes and norms were positively related to intentions with the former accounting for the most variance. Past sexting behaviour was found to explain additional variance in intentions such that having previously pressured a partner increased intentions to engage in the same behaviour in the future. Sexual violence prevention efforts may benefit from targeting young adults' attitudes and perceived social acceptance of coerced sexting, even in the context of a romantic relationship.

Michaud, Nicole

TALK TITLE: Altered Size Discrimination Following Optogenetic Modulation of Somatostatin Interneurons in Mouse Primary Visual Cortex

AUTHORS: Michaud, N.M., Boone, E.H., Long, C.R., Crowder, N.A.

ABSTRACT: Mice are an important animal model in vision research attributed to the availability of genetic tools used to explore neural circuits within the brain. In particular, optogenetic tools that use light to modify the activity of specific neural subtypes are useful for investigating the link between neural circuits and perception. Previous *in vitro* findings have implicated somatostatin expressing (SOM+) interneurons in visual surround suppression, which is a receptive field property proposed to underly the perception of object size or the boundaries between objects. Using transgenic mice that express a suppressive optogenetic protein (ArchT) in their SOM+ interneurons, we tested whether size perception depends on SOM+ activity within primary visual cortex. Our mice rapidly learned a two-alternative forced choice size discrimination task, and preliminary data with optogenetic inhibition of SOM+ interneurons showed altered psychometric functions indicating mice made more errors during difficult discriminations.

Monchesky, Ainsley

TALK TITLE: Comparing methodologies: Quantifying beak colouration change in photostimulated European starlings (*Sturnus vulgaris*)

AUTHORS: Ainsley N. Monchesky*, Broderick M. B. Parks, Leslie S. Phillmore

*Presenting author

ABSTRACT: Day length, or photoperiod, is the primary cue that signals breeding season in temperate songbirds. As day length increases (i.e., in spring), birds become photostimulated, which initiates a cascade of physiological changes to prepare birds for breeding. In European starlings (*Sturnus vulgaris*), one physiological change apparent in breeding season is a change in beak colouration; beaks change from black in winter to yellow in spring. There are multiple methods to quantify beak colouration in starlings, however inconsistencies among scales make comparability across the literature difficult. Here, we captured male and female starlings ($n=11$) from the wild in January when in non-breeding condition and housed them indoors on a long day photoperiod for 21 days to induce photostimulation. We photographed beaks throughout the experiment and scored beak colouration using three standard categorical scales used previously in the literature and using a new, continuous scale which quantified beak colouration change as a proportion. The continuous scale revealed changes over the experiment undetectable by the categorical scales; we therefore recommend using this approach in future studies of starling reproductive physiology to ensure accurate, and more fine-grain, representation of beak colouration.

Mungall, Luke

TALK TITLE: Do strategic voters have a strategic personality? Examining the role of Machiavellianism in vote-switching

AUTHORS: Luke R. Mungall, Scott Pruyers, Julie Blais

ABSTRACT: Many citizens are “floating” voters, who switch their vote between different parties from one election to the next (Stiers & Dassonneville, 2019). There are many antecedents to vote-switching, but one explanation is individual differences in personality traits. For example, previous studies have consistently found that curious and open-minded people are more likely to engage in vote-switching (Bakker et al., 2016; Erisen & Blais, 2016). Theoretically, people with Machiavellian personality traits should also be vote-switchers, since their cold, calculating, and status-driven nature should lead them to break partisan ties and vote for candidates that have the best chance of winning. A recent study suggests that Machiavellianism is not associated with a history of vote switching, but this cross-sectional study relied on self-reports of past voting behaviour, where the reasons for vote-switching are unknown (Pruyers & Blais, 2022). To resolve these limitations, we examined whether receiving pre-election polling information increases strategic vote-switching among those with Machiavellian personality traits. Our conference paper will present the results of this experiment.

Pandey, Anjali

TALK TITLE: Investigating Semantic Encoding in the Item-method Paradigm

AUTHORS: Anjali Pandey, Tracy Taylor-Helmick

ABSTRACT: In the study phase of an item-method directed forgetting task, memory instructions presumably operate by promoting further rehearsal of to-be-remembered (TBR) items and limiting encoding of to-be-forgotten (TBF) items. Consequently, in the test phase that follows, memory for TBR items is better than for TBF items. While some evidence suggests that semantic encoding of TBR items benefits from this differential rehearsal, other evidence suggests that the semantic properties of study items are just as likely to be encoded following an instruction to remember as they are following an instruction to forget. We continue our investigation of the nature of semantic encoding in the item-method paradigm by asking 1) what type of semantic information is encoded following an instruction to remember and despite an instruction to forget and 2) how do factors at encoding and retrieval affect recognition memory performance for the semantic representations of TBR and TBF study items.

Raja, Momina

TALK TITLE: Modifiable Characteristics That Predict Outcomes in a Community-Based Intervention for Preschoolers with Autism Spectrum Disorder (ASD)

AUTHORS: Momina Raja, Melissa Ferland, Helen Flanagan, Dorothy Chitty, Nancy Garon, Megan Thomas, and Isabel M. Smith

ABSTRACT: Pivotal Response Treatment (PRT) is an effective early intervention for children with autism spectrum disorder (ASD), but children do not benefit equally. Study aims were to (1) determine profiles based on children’s baseline positive affect, imitation, and toy play; (2) examine associations between profiles and change in expressive communication (EC) following intervention. Preschoolers with ASD ($n = 260$), enrolled in a provincial PRT-based program, contributed data including key variables coded from video-recorded pre-intervention play interactions with a clinician ($n = 130$). Cognitive (IQ) data were available for a subset. Latent profile analysis identified two groups differing on positive affect ($F(1,128) = 6.791, p = 0.010, \eta^2 = 0.05$), imitation ($F(1,128) = 299.8, p < 0.001, \eta^2 = 0.70$) and toy play ($F(1,128) = 188.5, p < 0.001, \eta^2 = 0.60$). Multilevel modelling found that profile membership and IQ predicted post-treatment EC changes. Findings provide guidance for individualizing early autism treatment.

Shephard, Aaron

TALK TITLE: Motives for substance use in individuals who use alcohol, cannabis, and nicotine

AUTHORS: Aaron Shephard, Sherry Stewart, Igor Yakovenko, Sean Barrett

ABSTRACT: **Background:** Alcohol, cannabis, and nicotine are three of the most commonly used drugs, with all three being frequently used by the same individual. Substance use motives typically vary between users, but it is unclear whether these differences are dependent on substance type, level of dependence, or simply come down to individual differences. **Method:** 516 Canadian adults with past month use of alcohol, cannabis, and nicotine completed online surveys querying their substance use history, level of dependence, and motives for use of each substance. Linear mixed models were conducted for each motive, including substance and dependence on each substance as fixed factors, with participant ID (intercepts) being included as random effects. **Results:** Different substances were more frequently used for specific motives (i.e., cannabis for enhancement motives, alcohol for social motives, nicotine for sensorimotor motives). Higher levels of dependence correlated with greater endorsement of each motive. The majority of variance in each model was explained by individual differences. **Discussion:** Previous research has largely focused on one substance at a time, with motives being analyzed for one specific substance. This research shows that, while certain motives were endorsed more for certain substances, individual differences play the largest role in the endorsement of motives.

Stamp, Jennifer

TALK TITLE: AI as your TA, Using ChatGPT to Vet Exam Questions

AUTHORS: Jennifer Stamp

ABSTRACT: Maintaining academic integrity is difficult in large undergraduate classes. The student-instructor ratio makes it almost impossible to verify the identity of the person who completes assessments and AI tools like ChatGPT do an excellent job of writing essays. Most strategies to strengthen academic integrity on traditional, in-person, test-based assessments focus on policing cheating efforts when they happen. Although these strategies work reasonably well face-to-face, they are ineffective in an online testing environment. In this talk, I'll share some strategies to leverage ChatGPT to design meaningful, cheat-resistant questions for open-book, online exams in 3rd year class in our Psych/Neuro program (Hormones & Behaviour).

Stevens, Leanne

TALK TITLE: An Ongoing Journey: Indigenization in Introduction to Psychology

AUTHORS: Dr. Leanne Stevens and Jocelyn Paul

ABSTRACT: During this session, we will outline our ongoing journey aimed at weaving Indigenous knowledge throughout our first-year Introduction to Psychology open textbook – highlighting significant contributions, ways of knowing, and historical injustices, where appropriate. The new content spans many topics, relevant to Psychology and Neuroscience, including (but not limited to) spatial memory, Models of Needs (self-actualization), and the impact of intergenerational trauma on development. By outlining our approach – including the voices of those with lived experience and expertise, engaging students-as-partners, revisiting and revising learning outcomes – we hope to concrete ideas and pathways for creating meaningful learning opportunities in the classroom (and beyond). The textbook is still a work in progress, but our hope is that this remains a feature rather than a flaw. The open nature of the text allows us to respond to student inquiries, recent events, new discoveries, and ultimately to remain relevant to our current and future students.

Wildeboer, Emily

TALK TITLE: Understanding the Experiences of Parents Engaged as Partners in Co-Created Knowledge Mobilization Initiatives in Children's Pain

AUTHORS: Emily M. Wildeboer, Christine T. Chambers, Nicole E. MacKenzie, Perri R. Tutelman, Justine Dol, Isabel Jordan, Jennifer A. Parker, Jennifer Stinson, & Erica Ehm

ABSTRACT: Engagement of patients as members of research teams has been increasingly recognized as an important part of patient-centered health research. The aim of this study is to understand the experiences of patient partners who were engaged in two co-created KM initiatives in children's pain. Data were collected as part of two collaborative KM initiatives about children's pain: It Doesn't Have to Hurt and Kids Cancer Pain, both of which engaged patient partners. Following their engagement, partners took part in a structured interview about their experiences. Interviews from each initiative were recorded, transcribed, and analyzed using an inductive content analysis approach. A total of 18 patient partners (9 from each initiative) participated. Overall, partners described their experiences very favourably. Several categories were generated across the data, including: 1) positive outcomes for partners, 2) difference between researcher and partner perspectives based on unique experiences; 3) contributions of patient partners directed the process and outcomes of the research; and 4) necessity of multiple patient partners on research team. These results can inform the development of best practices in patient engagement. Specifically, they can inform recommendations on how to improve engagement within pediatric pain research and KM.

POSTER ABSTRACTS

(In Alphabetical Order of Presenter)

All Posters Will Be Presented From 12:30 pm to 2 pm in the Lilyan E. White Study Centre on the 4th floor.

Brown, Richard

POSTER TITLE: Revising the Hebb Synapse for the 21st Century.

AUTHORS: Richard E. Brown. Department of Psychology and Neuroscience. Dalhousie University, Halifax Nova Scotia, Canada B3H 4R2

ABSTRACT: In 1949, Donald O. Hebb developed his neuropsychological postulate, which involved three neural processes: synaptic modifications (i.e., the Hebb Synapse), the cell assembly, and the phase sequence. The Hebb synapse describes how activity at pre- and post-synaptic neurons modulates the strength of a synapse (and synaptic networks). The Hebb synapse theory was based on electrical synapses, but the discovery of synaptic plasticity at NMDA receptors and the discovery of long-term potentiation and long-term depression provided evidence for the chemical nature of the Hebb synapse. However, new findings suggest that the concept of the Hebb synapse needs revision. We propose a 'hepta-partite' model of the synapse to account for the role of astrocytes, oligodendrocytes, microglia, the extracellular matrix (ECM), and the neurovascular unit (NVU) in the regulation of synaptogenesis and modulation of synaptic activity/plasticity. We also re-examine Hebbian plasticity in electrical synapses and consider the role of the gut microbiome in synaptic plasticity, as well as genetic and epigenetic mechanisms. Based on this new information, we revise Hebbian theories of the synapse, the cell assembly, and phase sequence underlying learning and memory to reflect the current understanding of synaptic plasticity, while upholding the legacy of Donald Hebb.

De Paola, Andrea

POSTER TITLE: Brief Retinal Inactivation Promotes Recovery of Microtubule-Associated Protein 2 in an Animal Model of Amblyopia

AUTHORS: Andrea De Paola, Paige Lauwerijssen, Jaimie Gregor, Sammy Pham, Annika Setterington, Sophie Shepherd, Theron Forbes, Sarah Keyes, Riana Giboi, Casandra Hayer, Haleigh Bach, Sophie Orr, Olivia Moreau, Sami Haghpavar, Ana Patton, Mark Henneberry, Dr. Kevin Duffy.

ABSTRACT: The formation and function of neural circuitry relies on interactions between structurally dynamic neurons. Microtubule-associated protein 2 (MAP2) has previously been identified as a prospective determinant of cytoskeletal plasticity. In an animal model of amblyopia, a visual disorder resulting from obstructed visual experience in one eye, monocular deprivation (MD) elicits and atrophy of neurons and loss of MAP2 in layers of lateral geniculate nucleus (LGN) serving the deprived eye. In this study, we investigated whether brief inactivation of the non-deprived eye with microinjection of tetrodotoxin could remediate the atrophy of neuron somata and loss of MAP2 caused by a prior MD. Retinal inactivation of the non-deprived eye for 2 days recovered cell size of neurons in deprived LGN layers and restored immunolabeling of MAP2 to normal. These findings indicate that brief retinal inactivation of the dominant eye is an effective approach to promote recovery of MAP2 following a period of amblyogenic rearing.

Drake, Richard

POSTER TITLE: Qualifying the “quantroversy”: Dissecting the working memory training debate

AUTHORS: Richard S. Drake, Gail A. Eskes, & Raymond M. Klein

ABSTRACT: The cognitive enhancement potential of computerized, game-based working memory training tasks continues to attract researchers and investors alike. Laboratory and commercial endeavours of the past two decades have been both basic and applied, with goals ranging from understanding the mechanisms of learning and transfer to improving academic performance or reducing age-related cognitive decline. Hundreds of scholars have participated in the debate of whether working memory training is effective. Putting aside valid criticisms of the debated question itself (“effective” is not specific and is only meaningful when paired with clearly defined goals), there is evidence to support both sides. However, on the basis of our systematic review of the working memory training meta-analyses, we argue that the controversy has more to do with differences of interpretation, rather than numeric differences. We thus qualify the alleged quantitative controversy (or “quantroversy”) accordingly. Please come and provide input or debate us!

Elgallad, Joseph

POSTER TITLE: The Importance of Timing: Effect of Retinal Inactivation on Neural Activity in the LGN is Reduced with Age

AUTHORS: Joseph Elgallad, Mason Eisner, Nhu Y Pham, Sappho Thompson, Selena Benjamin, Emma Cook, William Hazelton, Hilary S Comier, Taunya M Pynn-Crowe and Kevin Duffy

ABSTRACT: Conventional therapy for amblyopia is limited by issues such as compliance with treatment and recurrence of the disorder. We tested a novel treatment for amblyopia that involves reversible inactivation of the retina with TTX. Brief retinal inactivation promotes recovery from the amblyogenic effects of monocular deprivation at ages previously thought impossible. We subsequently found that the effectiveness of TTX to elicit plasticity is reduced in older animals. The diminishing effectiveness of inactivation in mature animals may be

the consequence of a decreased ability for inactivation to reduce synaptic activity. In this study we tested the hypothesis that the impact of brief retinal inactivation on synaptic activity is reduced with age. We assessed immunolabeling for vesicular glutamate transporter 2 (VGLUT2) as a proxy for synaptic activity within the dorsal lateral geniculate nucleus. We found a progressive loss in the ability of retinal inactivation to reduce levels of VGLUT2. That inactivation in older animals does not appreciably reduce activity levels in the visual system implies that older animals subjected to inactivation treatment for amblyopia will show limited recovery in comparison to younger ages.

Feltmate, Brett

POSTER TITLE: The Mini-Countryman Vehicle and Stimulus-Response Compatibility.

AUTHORS: Fiona E.Y. Burns-Hemingway, Raymond M. Klein, Brett B.T. Feltmate

ABSTRACT: In the "Union Jack" style taillights employed by some Mini-Countryman models, signaling to turn involves illuminating an inward-facing arrow. We asked: what occurs in a driver's mind when observing a vehicle signaling a leftward turn, but via a right-facing arrow? When viewed through a stimulus-response (SR) compatibility lens, the incongruency of the signal's irrelevant spatial shape (i.e, right-facing arrow) and its relevant spatial location (i.e, left-located) ought to impair one's ability to efficiently interpret drivers' intentions. This concern is compounded by the use of arrows as directional markers in turn-signs on roads, presumably leading drivers to adopt an attentional set biasing them towards interpreting the shape of arrows. To test this, we had participants make keypress responses (left, right) indicating the task-relevant property of arrows when presented as turn-signs (shape, ignoring location), or turn-signals (location, ignoring shape) within a mock highway display. For turn-sign responses we replicated the classic Spatial-Stroop effect (performance costs due to incongruency between arrows' task-relevant shape and irrelevant location). For turn-signal responses, we observed a "converse" Spatial-Stroop effect wherein performance suffered when arrows' task-relevant location was incongruent with their irrelevant shape. These results suggest that the "Union Jack" style taillights may be detrimental to road safety.

Gendron, William

POSTER TITLE: Effects of Intranasal Insulin on Memory and Physiology in the 5xFAD Mouse Model of Alzheimer's Disease

AUTHORS: William H. Gendron¹, Emre Fertan¹, Kyle M Roddick¹, Younes Anini^{2,3}, Richard E. Brown^{1, 2}
Departments of ¹Psychology, ²Physiology and Biophysics, ³Obstetrics and gynecology. Dalhousie University, Halifax Nova Scotia Canada B3H 4R2

ABSTRACT: As well as a disorder of cognitive function, Alzheimer's Disease (AD) is a metabolic disorder related to insulin resistance in the brain and has been termed "type 3 diabetes". Intranasal insulin administration has been proposed as a treatment for AD. The 5xFAD mouse shows age-related decreases in body weight as well as cognitive and motor deficits. As defective insulin signaling in AD is an example of metabolic dysregulation, this study examined whether intranasal delivery of insulin (INDI) at low (LI; 0.875 U) and high (HI; 1.75 U) doses would ameliorate these deficits compared to saline (S) in 12-month-old female 5xFAD and WT (B6SJL) mice. The 5xFAD mice had poorer performance on the balance beam than WT mice ($p < 0.05$) and there was no effect of insulin. The HI group performed better than the saline group in the novel object recognition test ($p < 0.05$) and showed better memory performance in the probe trial of the Morris Water Maze test. Although INDI had no significant effect on body weights, HI doses of INDI increased the liver, spleen, and kidney weights in both WT

and 5xFAD mice ($p < 0.05$). Brown adipose tissue weights in the HI group were lower than in the S and LI groups. P-Akt signaling was increased by INDI in a dose-dependent manner ($p < 0.05$). These findings indicate that visuo-spatial memory and organ weights in female 5xFAD mice are affected by treatment with intra-nasal insulin in a dose dependent manner.

Glenham, Cecilia

POSTER TITLE: Grooming Behaviour in The Neurexin1+/- Mouse Model of Autism Spectrum Disorder

AUTHORS: Cecilia Glenham, Dr. Richard Brown

ABSTRACT: Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder characterized by deficits in communication, social interaction, and repetitive behaviors. Mutations of the Neurexin (NRXN) gene result in disrupted synaptic function and have been strongly implicated in ASD, although the precise mechanisms remain unclear. Mouse self-grooming is a repetitive, sequentially patterned behavior model, making it an ideal translational tool for ASD research. The Neurexin1+/- (Nrnx1+/-) is a novel mouse model of ASD, and our study was the first to perform grooming research for this strain. In our study, we examined grooming microstructure in male and female Nrnx1+/- mice compared to wild-type (Nrnx1+/+) mice by using quantitative and sequential analysis. Our research did not yield any statistically significant results, however, some interesting non-significant sex differences were observed. Our research emphasizes the importance of investigating specific behavioural microstructure, with particular attention to potential sexually dimorphic differences. Additionally, we highlight the need for further research in Nrnx1+/- mice.

Hartley, Mackenzie

POSTER TITLE: Naturally-Formed Social Hierarchies are not Associated with Indicators of Stress and Anxiety in Male Mice

AUTHORS: Mackenzie Hartley, Millie Bhaskara, Alex MacKay, Tamara Franklin

ABSTRACT: Within social dominance hierarchies, lower social rank is associated with higher levels of stress. Hdac2 is an epigenetic protein regulated by the glucocorticoid receptors activated during the stress response and, therefore, shows greater expression in individuals with increased stress levels. This study aimed to provide evidence to correlate stress, anxiety, and Hdac2 protein expression in the hippocampus with social rank in male mice. We performed western blots to measure Hdac2 protein expression, used adrenal gland weights to measure stress and a triple-test paradigm to measure anxiety. We expected to find increased stress, anxiety, and Hdac2 protein expression displayed by lower-ranked individuals, however, our findings suggest that there is no link between stress, anxiety, Hdac2 protein expression, and social rank in hierarchies naturally formed by male mice. These findings may be due to the absence of resource competition in lab conditions, or due to stress induced by the behavioural testing procedure.

Hayward, Frances

POSTER TITLE: Investigating the Effect of Acute Psychosocial Stress and Aerobic Exercise on Salivary Cortisol Concentrations in Young Adults

AUTHORS: Frances R. Hayward (others listed on poster: Sophie Inkpen, Elizabeth M. Myles, Cory Munroe, Heather F. Neyedlie, Tara S. Perrot)

ABSTRACT: When humans are stressed, they respond predominately by activation of their hypothalamic-pituitary adrenal (HPA) axis. The HPA axis releases cortisol, a glucocorticoid; crucial in the moment but culprit to the long-term damage caused by stress. Stress related disorders are mitigated by aerobic exercise (AE), which, if done prior to psychosocial stress exposure, lessens cortisol response. AE, a physical stressor, is thought to accomplish this by causing early activation of the HPA axis' regulating mechanism: glucocorticoid negative feedback. The present study aimed to investigate if this impact on cortisol was still present if the order was reversed, and the AE was performed after exposure to psychosocial stress. We hoped to provide evidence that AE might be capable of mitigating negative aspects of the stress response retroactively. Three experimental conditions were conducted, each with four participants; the first group participated only in a psychosocial stress paradigm (The Montreal Imaging Stress Task; Dedovic et al., 2005), the second group participated only in moderately intense AE, and the third group participated in a psychosocial stress paradigm, followed by moderate AE. Six saliva samples were collected throughout the sessions and analyzed later using an enzyme-linked immunosorbent assay. A repeated measures ANOVA revealed that cortisol levels were highest at the end of the session in those who had participated in both the stress and AE paradigms. Contrary to what we had imagined, these results indicate a potential conflation of psychosocial stress and AE when performed in this order, resulting in a subsequent increase of HPA axis activation.

Jacques, Sophie

POSTER TITLE: Language and Executive Function in Dogs: Testing for the Universality of the Relation

AUTHORS: Sophie Jacques, Catherine Reeve, Lara Presser & Karina Alvarado Ochoa

ABSTRACT: Language predicts executive function (EF) in humans (Müller et al., 2009). Cross-species comparisons examining this relation can help determine the universality of this relation. Dogs are ideally suited for this, as owners direct and adapt their speech to dogs (Mitchell, 2001) and dogs respond systematically to words, particularly action commands (Mill, 2005; Ramos & Mills, 2019), demonstrating some receptive abilities, although they vary widely in their reported receptive abilities (Reeve & Jacques, 2022). Dogs also possess rudimentary EFs, measured using owner questionnaires (Foraita et al., 2022) and behavioural tests (Tiira et al., 2020). We examined relations between language and EF in 104 dogs (54 females; Mage=5.21yrs) using Reeve and Jacques' (2022) owner vocabulary checklist, with Total Vocabulary and Action Commands as scores. EF was assessed using an incrementally difficult behavioural search task and a 12-item owner questionnaire. Even after controlling for dogs' age and sex, Total Vocabulary and Action Commands correlated with search task performance and owner responses on the EF questionnaire, indicating that the relation between language and EF extends to another species. Understanding inter-species similarities and differences can help shed light into the nature of the relation between these abilities.

Killeen, Jack

POSTER TITLE: DRIFT-OFF: Diabetes Related Insomnia in Families and Teenagers – Optimising control and Facing Fears

AUTHORS: Jack Killeen, Department of Psychology and Neuroscience, Dalhousie University, Halifax, NS, IWK Health Centre, Halifax, NS

Dr. Megan Thomas, Department of Psychology and Neuroscience and Associate Professor in Department of Pediatrics, Dalhousie University, Halifax, NS, IWK Health Centre, Halifax, NS, Lancaster University, Lancaster, UK

Dr. Teresa Pinto, Associate Professor, Department of Pediatrics, Dalhousie University, Halifax, NS, IWK Health

Centre, Halifax, NS

ABSTRACT: Type 1 diabetes mellitus (T1D) is a chronic condition that occurs when the pancreas makes little to no insulin, thereby causing raised levels of blood glucose. Sleep disturbances are common in people with T1D due to symptoms of high or low blood glucose levels, anxiety around glucose levels and monitoring throughout the night. Current research focuses on sleep disturbances in adults with T1D, with less known about the relationship in younger populations. This correlational, mixed methods, exploratory study, will use questionnaires, a 7-day sleep diary, and metabolic parameters to focus on the relationship between glycaemic control and sleep in children and adolescent patients and their parents/caregivers. A maximum of 50 children/adolescent and parent/caregiver pairings will be recruited through a tertiary children's hospital diabetes clinic. We hypothesize that poorer glycaemic outcomes will be related to more disrupted sleep in both children/adolescents and parents/caregivers. We also hypothesize that the use of diabetes related technologies, such as Continuous Glucose Monitors (CGM), will decrease Fear of Hypoglycaemia (FoH), the anxiety surrounding low blood glucose levels and its symptoms, in both children/adolescents with CGMs and their parents/caregivers. The results of this study may influence sleep management recommendations for children and adolescents with T1D and their caregivers. Our aim is to create a familial view of diabetes related sleep differences and stress in the home and empower patients and their families to make informed choices to optimize their health. Promoting healthy sleep is an important public health measure, especially in children/adolescents with diabetes as T1D already has widespread negative effects on an individual's health and well-being.

MacDougall, Blythe

POSTER TITLE: Examining the Barriers to and Experiences of Nature Exposure in Undergraduate Students

AUTHORS: Blythe MacDougall, Rachel McMillan, Yasmeeen Ibrahim, George Fazaa, & Shannon A. Johnson

ABSTRACT: Undergraduate student mental health is a rising concern and mental health services for undergraduates are limited. Accessible, affordable, and effective treatments and preventative approaches are needed to address these challenges. Previous research has demonstrated nature exposure's benefits on mental health. We investigated the barriers and experiences of nature exposure in undergraduate students. Eighty-six undergraduate students completed a novel intervention, "Nature Matters". This intervention, designed to increase exposure to and connection with nature, includes two components: seven short videos, followed by a written companion. Consistent with prior research and our hypotheses, undergraduates identified key barriers to nature exposure, including a lack of spare time, insufficient knowledge on how to access nature, and unfavourable weather conditions. As expected, inductive thematic analysis of participants' reflections before and after the brief virtual nature exposure revealed that the video positively affected perceptions of their current states. The nature exposure evoked positive changes in negative state perceptions with most responses suggesting virtual nature exposure was both an effective and pleasant experience.

Myatt, Amber

POSTER TITLE: Employing the RE-AIM framework to assess the implementation success of Better Nights, Better Days during the COVID-19 pandemic

AUTHORS: Myatt, A.1 (Submitting/Presenting Author), Keys, E.2, & Corkum, P.1

1 Dalhousie University, 2 University of British Columbia Okanagan

ABSTRACT: Background. To address worsening sleep problems in children during the pandemic, an existing eHealth program for sleep problems in children, Better Nights, Better Days (BNBD), was modified to be appropriate in the pandemic context. This study used the Reach, Effectiveness, Adoption, Implementation, Maintenance (RE-AIM) framework to understand how families of children with behavioural sleep problems implemented the program in the context of COVID-19. **Methods.** Implementation was assessed based on: the best way to reach parents during COVID-19 (Reach); the effectiveness of the program in terms of child sleep and family resilience (Effectiveness); how the program was adopted within families (Adoption); the percentage of parents who accessed and/or completed the program (Implementation); and if parents maintained use of program strategies after program completion (Maintenance). **Results.** Results suggest that most parents were reached through online advertisements (74.2%) but that many parents who enrolled in the program did not end up accessing it (70.5%) or completing a clinically significant dose (84.5%). However, those that did complete the program had high fidelity with implementing the strategies and the program was just as effective as pre-pandemic for families who completed it.

Ortibus, Wyatt

POSTER TITLE: Olfactory Discrimination Tasks and Reversal Learning in a Neurexin1 (+/-) mouse model of Autism Spectrum Disorder

AUTHORS: Wyatt Ortibus, Dr. Kyle Roddick & Dr. Richard Brown

ABSTRACT: The Nrnx1(Nrnx1 +/-) mouse model is a novel strain with no previous research into their learning and memory performance. Neurexins (NRXN) are a family of type-1 membrane protein synaptic organizers that bind neuroligins to form Ca (2+) dependent complexes that shape synapse properties of neural networks by synapse specificity. Synaptic function is critical for learning and memory processes and Neurexins are essential for synaptic function as alterations to Neurexins are implicated in neurodevelopmental disorders (autism and schizophrenia). Our experiment examines the effects of Neurexin mutations on learning and memory using transgenic mice and operant conditioning tasks using an olfactometer. Using an olfactometer Nrnx1 (+/-) mice and wildtype (Nrnx1 +/+) littermates we scored total errors during three odor discrimination learning tasks. The Nrnx1 (+/-) and wildtype controls did not differ in their total errors to 85% accuracy criterion in our Introduction and New Operant Odor Pair memory tasks, indicating that both genotypes are effectively able acquire novel operant pairs. Our results show significantly more errors on reversal learning in the Nrnx1 (+/-) mice compared to wildtype mice as the reversal learning task requires more cognitive flexibility than the prior learning assessments. The Nrnx1+/- mice underperform in reversal learning olfaction tasks within an olfactometer due to the dysregulation of synaptic function compared to controls.

Oulton, Madison

POSTER TITLE: Effects of Sleep Deprivation on the Density and Size of Mitochondria in Axon Terminals Presynaptic to Rat Orexin Neurons

AUTHORS: Madison Oulton, Tatjana Golovin & Kazue Semba

ABSTRACT: Orexin (ORX) neurons are wake-active and wake-promoting neurons found in the lateral hypothalamus. During sleep deprivation (SD), we would expect not just ORX neurons but their excitatory inputs to be active. Considering that these inputs utilize their cellular energy stores to fuel synaptic activities, we would expect the mitochondria within these terminals to increase in size and density after SD. To test this hypothesis, adult male Wistar rats were either sleep deprived (n=3) or left undisturbed (n=3) for 6 hours during the first half

of the light phase, then immediately sacrificed. Using correlative light-electron microscopy, several ORX neurons were identified and imaged in serial sections at 30-50 nm thickness. The number and length of mitochondria per axon terminal for several hundreds of axon terminals presynaptic to 6 ORX neurons were analyzed. Although the author is required to remain blind to the experimental condition in order to continue to work on the project, significant differences between individual animals were found, potentially supporting the hypothesis that mitochondria in axon terminals presynaptic to ORX neurons increase in size and density after SD.

Smith, Seth

POSTER TITLE: Does early ineffective occlusion treatment of amblyopia in kittens reduce the effectiveness of later longer treatments that are typically quite successful?

AUTHORS: Seth Smith, Kevin Duffy & Donald Mitchell

ABSTRACT: A smattering of reports on patching therapy (i.e. occlusion of the “good” eye) for amblyopia in children suggest that prior unsuccessful patching treatment may reduce the efficacy of subsequent longer therapy. We are investigating this possibility in kittens that were rendered amblyopic in one eye by monocular occlusion for 7 days starting at P21 days of age. Subsequently, two groups of kittens receive 3 weeks of occlusion of the fellow (“good”) eye that starts at 5 weeks of age but in one group is preceded by 3 days of fellow-eye occlusion beginning at the end of the initial period of monocular occlusion at P28. Preliminary behavioural measurements of the grating visual acuity of each eye suggest that the early short period of occlusion of the fellow eye reduces the efficacy of the second long period of occlusion of this eye.

Yazbek, Yara

POSTER TITLE: The Impact of Accented Prosocial Behaviour on Preschoolers’ Selective Word Learning

AUTHORS: Yara Yazbek, Madison MacLachlan, Rebeka Workye, Emma Tupper, & Dr. Drew Weatherhead

ABSTRACT: Children’s language development involves learning from others to navigate linguistic variations such as native and non-native accented speech. Social factors like nationality and accents, as well as behavioural traits play a role in word learning. We investigate how prosocial behaviour interacts with accented information to make an unfamiliar accented speaker more preferred through a label endorsement task. It was hypothesized that preschoolers are more likely to endorse a label presented by a non-native accented speaker when receiving prosocial information about their social group. Analyses show no significant differences in the proportion of times the Wuppit label was chosen between the prosocial, antisocial, and neutral conditions. Post-hoc comparisons showed slight but insignificant differences, but the low-powered sample size prevented statistical significance. The results of this study highlight the need for further research investigating these factors, as it can have implications for the development of programs that promote positive representations of minority language groups.

Yorukoglu, Sare

POSTER TITLE: Cultural Adaptations of Developmental Interventions (DIs) and Naturalistic Developmental Behavioural Interventions (NDBIs) for Preschoolers with Autism Spectrum Disorder: A Delphi Study

AUTHORS: Sare Yorukoglu Supervisors: Isabel Smith & Teba Hamodat

ABSTRACT: **Background:** Developmental Interventions (DIs) and Naturalistic Developmental Behavioural Interventions (NDBIs) designed for pre-schoolers with Autism Spectrum Disorder are accepted as the predominant treatment methods that target social processes and emotional development. Although there is existing literature on how both interventions might take form in various ethnic/geographical/cultural settings, most researchers have yet to describe the adaptation process they have taken to adjust interventions. **Objective:** This study aims to address which modifications and adaptations to DIs and NDBIs would be the most suitable for preschoolers with ASD from different ethnic/ geographical/cultural settings. **Methods:** We are conducting a Delphi study where experts in the ASD field with related experiences will be asked questions based on the core elements of the Framework for Reporting Adaptations and Modifications-Expanded (FRAME) (Stirman et al., 2019). Twenty experts agreed to partake in the study. The rounds of questionnaires will continue until a consensus is reached. **Preliminary Findings:** After the completion of the first round, we conducted a content analysis based on Graneheim & Lundman's framework (2003) to analyze the qualitative data. The analysis has resulted in the identification of over 100 condensed meaning units, which will be presented to the expert panel in the second round. The experts will be asked to rate their agreement on each unit using a 6-point Likert scale, enabling the prioritization and refinement of the adaptation. **Significance:** The study's findings will address the critical need for culturally sensitive adaptations of developmental interventions for preschoolers with Autism Spectrum Disorder (ASD).